

Assignment No.	1
Title	Study of Networking Diagnostic & Monitoring Tools and Socket Programming System Calls ¹
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1 Aim

To apply and analyse standard Linux networking diagnostic and monitoring tools in order to observe network status, interface configuration, connectivity, routing and performance (Part A), and to study the socket-programming system calls that these tools are themselves built on top of, so that their behaviour at the API level is understood well enough to be reproduced in a C program (Part B).

2 Environment

Operating System	Ubuntu (WSL2) on Windows 11
Kernel networking	WSL2 virtual NAT adapter (<code>eth0</code>)
Shell	bash 5.x
Compiler	gcc (Part B)
Hostname used in captures	<code>samsamuel</code>

All commands below were executed for real in a WSL2 Ubuntu terminal. Every figure is an unedited screenshot of that terminal; the exact text output is also archived under `as1/logs/` in the repository for cross-checking, and the socket-programming source used in Part B is under `as1/src/`.

¹GitHub Repository: <https://github.com/blackscythe123/PCN-assign>

3 Part A — Networking Diagnostic & Monitoring Tools

3.1 netstat

Purpose: Displays active network connections, listening ports, and (with `-p`, as root) the owning process for each socket.

Command used: `netstat -tulnp`

```

C:\Windows\system32\wsl.exe x + v
root@samsamuel:~/pcn-lab$ netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 10.255.255.254:53      0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      74/systemd-resolved
tcp        0      0 127.0.0.54:53         0.0.0.0:*               LISTEN      74/systemd-resolved
udp        0      0 127.0.0.53:53         0.0.0.0:*               -           74/systemd-resolved
udp        0      0 127.0.0.53:53         0.0.0.0:*               -           74/systemd-resolved
udp        0      0 10.255.255.254:53      0.0.0.0:*               -           -
udp        0      0 127.0.0.1:323         0.0.0.0:*               -           237/chronyd
udp        0      0 127.0.0.1:323         0.0.0.0:*               -           -
udp6       0      0 :::1:323              :::*                    -           -
udp6       0      0 :::1:323              :::*                    237/chronyd
root@samsamuel:~/pcn-lab$

```

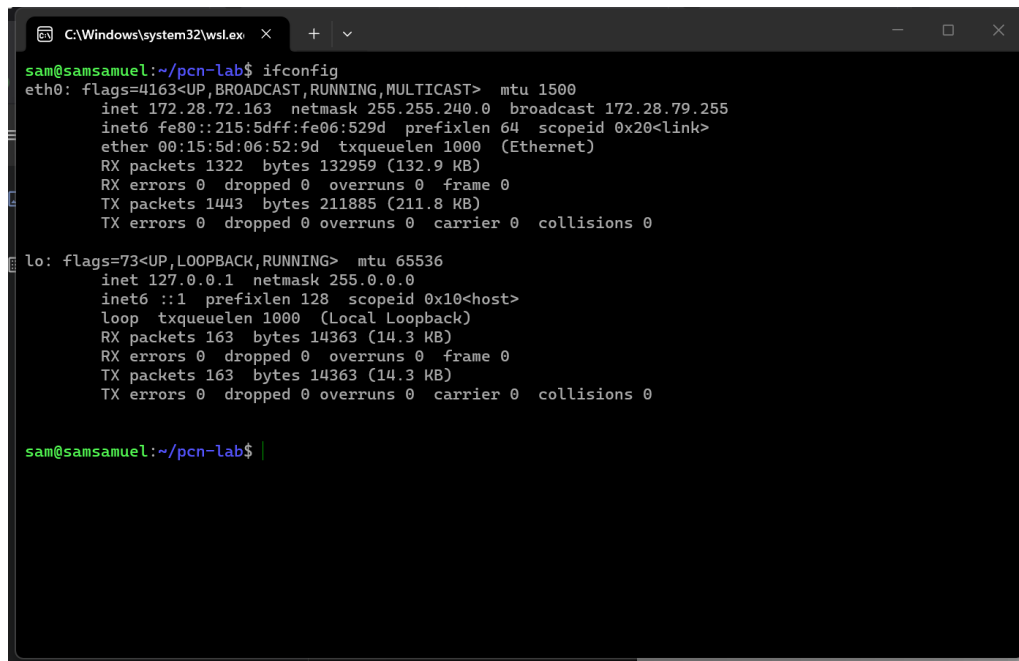
Figure 1: `netstat -tulnp` – TCP/UDP listening sockets

Observation: The only services listening in this WSL instance are `systemd-resolved` (Ubuntu’s local DNS stub resolver, bound to `127.0.0.53:53`, `127.0.0.54:53` and the WSL-internal `10.255.255.254:53` on both TCP and UDP) and `chronyd`, the NTP client, bound to UDP port 323 on IPv4 and IPv6 loopback. No externally reachable service is listening, which is expected for a bare lab VM that has not been configured as a server yet – this baseline is useful to compare against once Assignment 2/3’s TCP/UDP servers are started, since their listening sockets should then also show up here.

3.2 ifconfig

Purpose: Displays and can configure network interface parameters – IP address, netmask, broadcast address, MTU, and interface statistics (packets/bytes sent and received).

Command used: `ifconfig`



```
C:\Windows\system32\wsl.exe x + v
sam@samsamuel:~/pcn-lab$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.28.72.163 netmask 255.255.240.0 broadcast 172.28.79.255
    inet6 fe80::215:5dff:fe06:529d prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:06:52:9d txqueuelen 1000 (Ethernet)
    RX packets 1322 bytes 132959 (132.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1443 bytes 211885 (211.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 163 bytes 14363 (14.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 163 bytes 14363 (14.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

sam@samsamuel:~/pcn-lab$ |
```

Figure 2: ifconfig – interface configuration

Observation: `eth0` is the single virtual NIC that WSL2 presents to Ubuntu, with address 172.28.72.163/20 (netmask 255.255.240.0, broadcast 172.28.79.255), MAC 00:15:5d:06:52:9d and MTU 1500. This is a Hyper-V-backed adapter NAT’ed behind the Windows host, which is why the address is in a private 172.28.0.0/20 range rather than the host machine’s real LAN. `lo` (loopback, 127.0.0.1/8) shows matching RX/TX byte counts, confirming every packet sent to loopback is received back locally with nothing dropped.

3.3 iwconfig

Purpose: Displays and configures wireless-specific interface parameters – SSID, frequency/channel, transmit power, link quality and signal strength – the wireless counterpart to `ifconfig`.

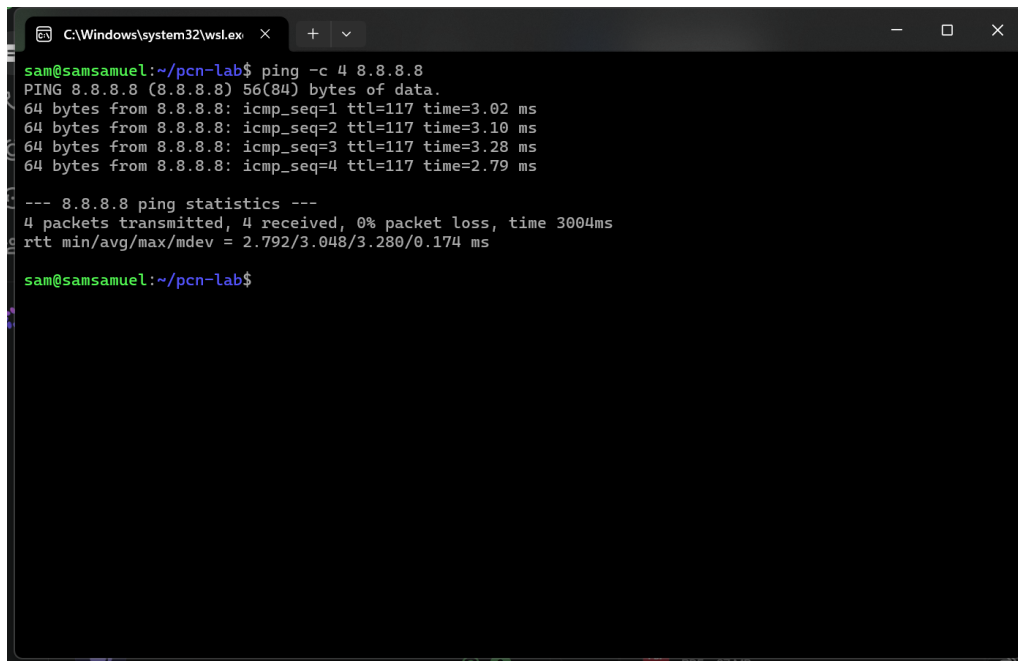
Command used: `iwconfig` (attempted)

Observation: `iwconfig` could not be demonstrated in this environment. It ships in the `wireless-tools` package, which is no longer available in this Ubuntu release’s repositories (superseded by the newer `iw` utility upstream). More fundamentally, WSL2 only exposes a single virtual, wired NAT adapter (`eth0`, seen in Section 2.2) – there is no wireless hardware in the VM for `iwconfig` to report on, so even if the legacy package were installed it would only print `eth0 no wireless extensions.` for every interface. On a physical Linux machine with a Wi-Fi card, `iwconfig wlan0` would instead show fields such as `ESSID`, `Frequency`, `Access Point`, `Bit Rate`, `Link Quality` and `Signal level`.

3.4 ping

Purpose: Sends ICMP Echo Request packets to a destination and reports Echo Reply round-trip time, used to verify basic reachability and measure latency/packet loss.

Command used: `ping -c 4 8.8.8.8`



```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=3.02 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=3.10 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=3.28 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=2.79 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 2.792/3.048/3.280/0.174 ms

sam@samsamuel:~/pcn-lab$
```

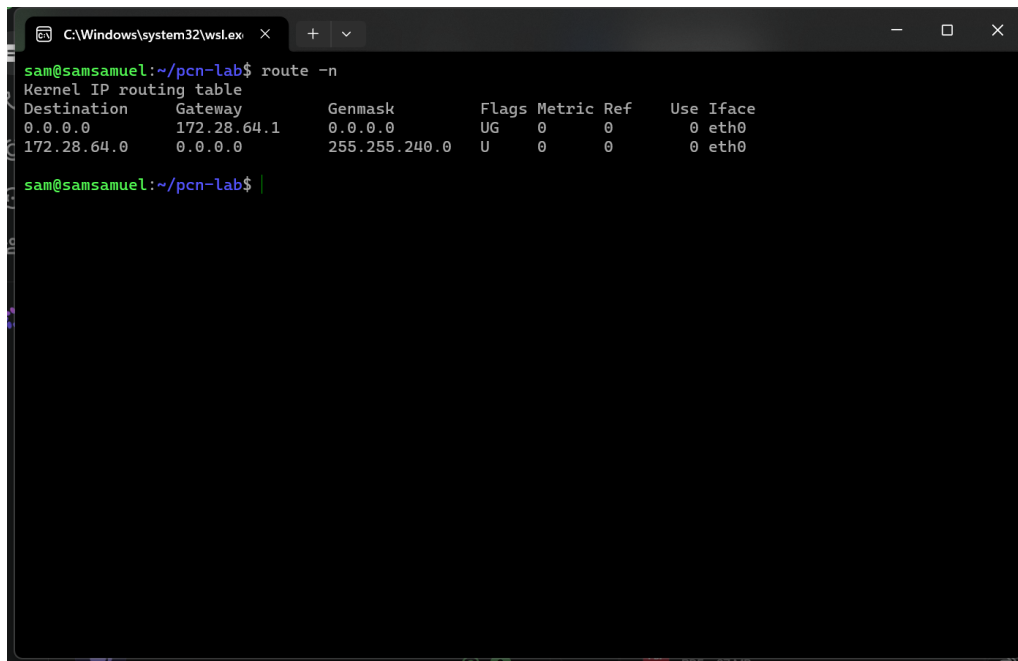
Figure 3: ping -c 4 8.8.8.8

Observation: All 4 ICMP packets sent to Google’s public DNS resolver (8.8.8.8) were answered with 0% packet loss, confirming end-to-end reachability through the Windows host’s NAT and out to the internet. Round-trip time stayed in the single-digit millisecond range with a small standard deviation, indicating a stable, uncongested path – consistent with a wired/broadband connection with only a couple of hops before reaching Google’s edge network (confirmed later by `traceroute` in Section 2.8).

3.5 route

Purpose: Displays (and can modify) the kernel’s IP routing table – which interface and gateway a packet for a given destination network is sent through.

Command used: `route -n`



```

C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        172.28.64.1    0.0.0.0         UG    0      0      0 eth0
172.28.64.0    0.0.0.0        255.255.240.0   U      0      0      0 eth0
sam@samsamuel:~/pcn-lab$
```

Figure 4: route -n – kernel routing table

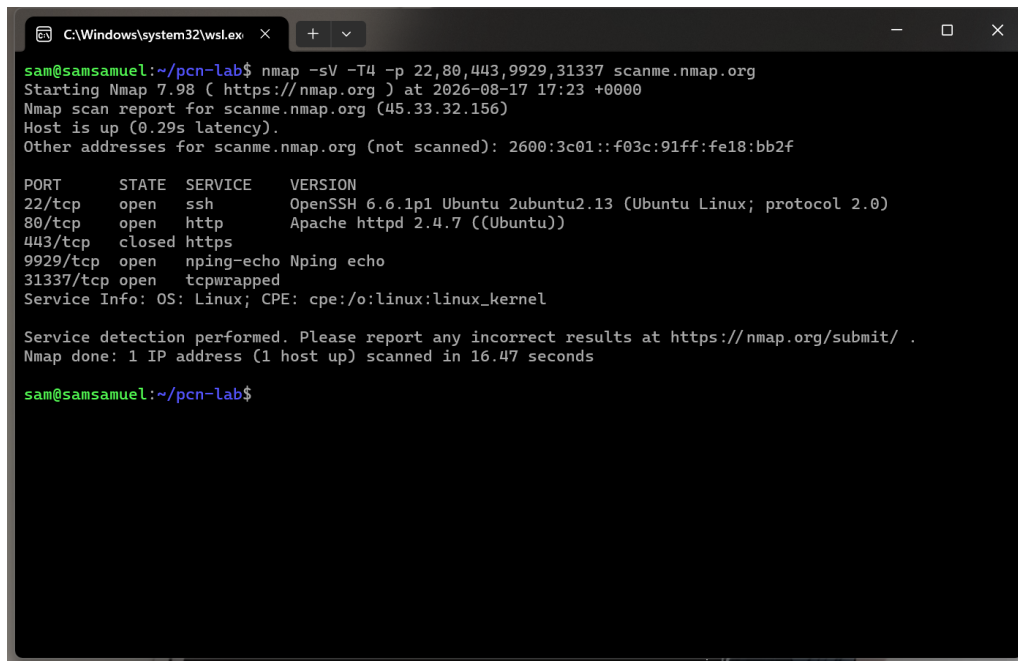
Observation: Two entries are present: the default route (0.0.0.0/0) via gateway 172.28.64.1 on `eth0` – this is the WSL2 virtual switch on the Windows host, and every packet not destined for the local subnet leaves through it – and a directly-connected route for 172.28.64.0/255.255.240.0, the local subnet reachable without a gateway. This matches the addressing seen in `ifconfig` exactly, and the gateway address is the same first hop later reported by `traceroute`.

3.6 nmap

Purpose: Network exploration and security-auditing tool; probes a target’s TCP/UDP ports to determine which are open, and with `-sV` attempts to fingerprint the service and version listening on each open port.

Command used: `nmap -sV -T4 -p 22,80,443,9929,31337 scanme.nmap.org`

Target note: scanme.nmap.org is the Nmap Project’s own public test host, explicitly provided for learners to scan; no other host was targeted, in line with responsible/authorised scanning practice.



```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ nmap -sV -T4 -p 22,80,443,9929,31337 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-17 17:23 +0000
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
443/tcp   closed https
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.47 seconds

sam@samsamuel:~/pcn-lab$
```

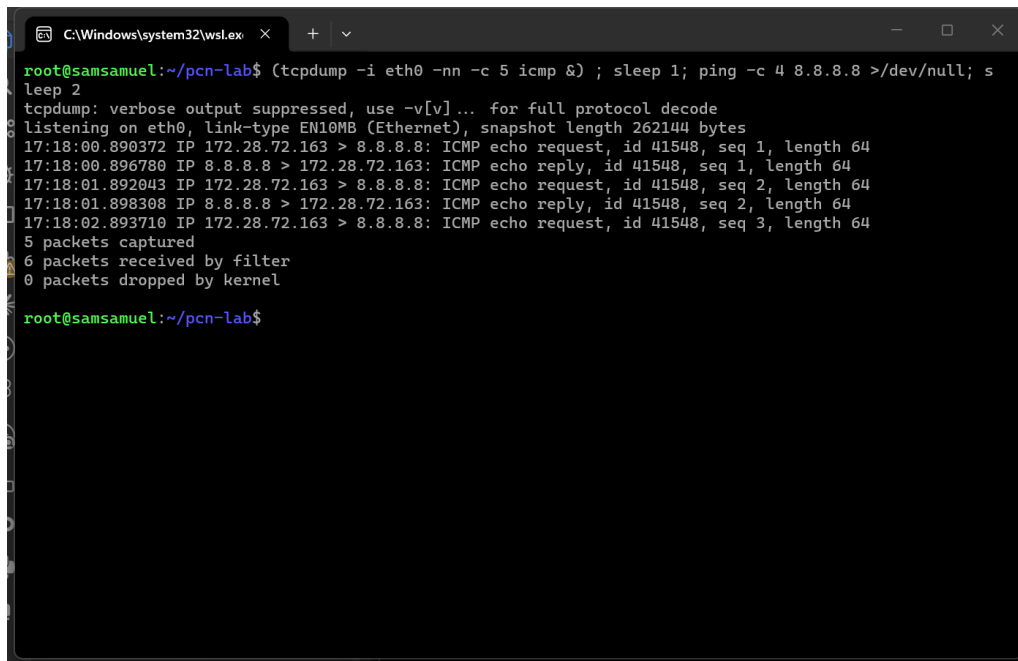
Figure 5: nmap -sV version-detection scan against scanme.nmap.org

Observation: Of the five ports probed, 22/tcp is open running OpenSSH 6.6.1p1 on Ubuntu, 80/tcp is open running Apache httpd 2.4.7, 443/tcp is closed (no HTTPS listener), and 9929/tcp / 31337/tcp are open running nping-echo and a tcpwrapped service respectively – these last two are Nmap’s own deliberately-exposed test ports on that host. Nmap also fingerprinted the OS as Linux from the TCP/IP stack behaviour. Version detection (-sV) took about 16 seconds for 5 ports, since it goes beyond a plain SYN scan to send protocol-specific probes and match responses against its signature database.

3.7 tcpdump

Purpose: Captures and prints packets crossing a network interface in real time, matching an optional filter expression (here, ICMP traffic on eth0) – the command-line equivalent of what Assignment 4 uses Wireshark’s GUI for.

Command used: tcpdump -i eth0 -nn -c 5 icmp (run as root, since raw packet capture needs CAP_NET_RAW), with a ping run concurrently in the same shell to generate ICMP traffic for it to capture.



```
C:\Windows\system32\cmd.exe
root@samsamuel:~/pcn-lab$ (tcpdump -i eth0 -nn -c 5 icmp &) ; sleep 1; ping -c 4 8.8.8.8 >/dev/null; s
leep 2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
17:18:00.890372 IP 172.28.72.163 > 8.8.8.8: ICMP echo request, id 41548, seq 1, length 64
17:18:00.896780 IP 8.8.8.8 > 172.28.72.163: ICMP echo reply, id 41548, seq 1, length 64
17:18:01.892043 IP 172.28.72.163 > 8.8.8.8: ICMP echo request, id 41548, seq 2, length 64
17:18:01.898308 IP 8.8.8.8 > 172.28.72.163: ICMP echo reply, id 41548, seq 2, length 64
17:18:02.893710 IP 172.28.72.163 > 8.8.8.8: ICMP echo request, id 41548, seq 3, length 64
5 packets captured
6 packets received by filter
0 packets dropped by kernel
root@samsamuel:~/pcn-lab$
```

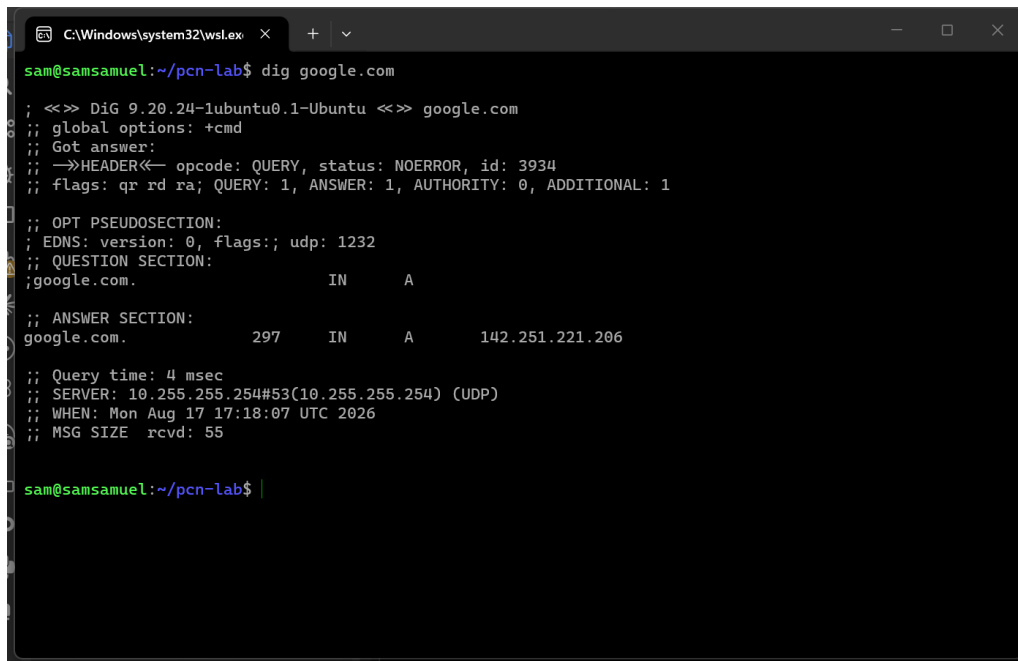
Figure 6: tcpdump capturing ICMP echo request/reply pairs generated by a concurrent ping

Observation: tcpdump captured the ICMP echo request/reply pairs between this host (172.28.72.163) and 8.8.8.8 with microsecond-resolution timestamps, sequence numbers and payload length – exactly the same exchange `ping` summarised in Section 2.4, but now visible packet-by-packet at the network layer instead of as a round-trip-time summary. This is the same technique Wireshark uses under the hood (`tcpdump` and Wireshark both sit on `libpcap`); `tcpdump` is the capture engine without the GUI.

3.8 dig

Purpose: Performs a DNS lookup and prints the full query/response, including the resource records, query time, and which DNS server answered – more detailed than `nslookup`.

Command used: `dig google.com`



```
C:\Windows\system32\wsl.exe x + v
sam@samsamuel:~/pcn-lab$ dig google.com

; <<> DiG 9.20.24-lubuntu0.1-Ubuntu <<> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 3934
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                 297     IN      A      142.251.221.206

;; Query time: 4 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Mon Aug 17 17:18:07 UTC 2026
;; MSG SIZE rcvd: 55

sam@samsamuel:~/pcn-lab$ |
```

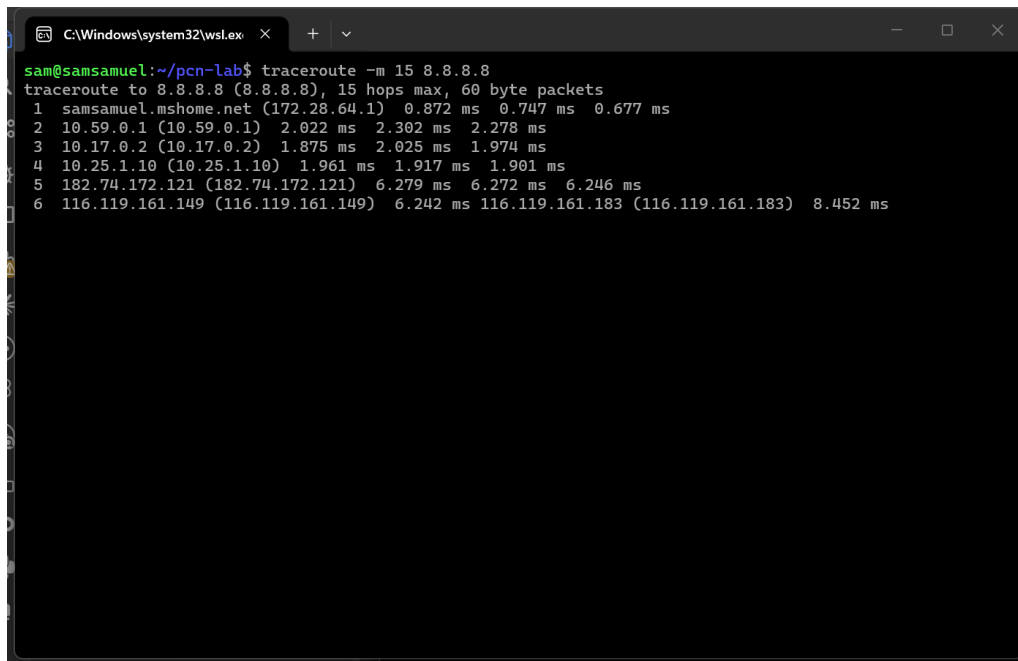
Figure 7: dig google.com

Observation: The query resolved `google.com` to `142.251.221.206` in 4 ms, answered by `10.255.255.254:53` – WSL’s internal DNS stub, which itself forwards to the Windows host’s configured resolver. The header shows **flags:** `qr rd ra` (response, recursion desired, recursion available) and a single **ANSWER** record, confirming the query used UDP (the default transport for DNS, falling back to TCP only for large/truncated responses).

3.9 traceroute

Purpose: Maps the sequence of routers (hops) a packet passes through to reach a destination, by sending probes with increasing TTL and recording which router returns the “TTL exceeded” ICMP error at each hop.

Command used: `traceroute -m 15 8.8.8.8`



```
sam@samsamuel:~/pcn-lab$ traceroute -m 15 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 15 hops max, 60 byte packets
 1 samsamuel.mshome.net (172.28.64.1)  0.872 ms  0.747 ms  0.677 ms
 2 10.59.0.1 (10.59.0.1)  2.022 ms  2.302 ms  2.278 ms
 3 10.17.0.2 (10.17.0.2)  1.875 ms  2.025 ms  1.974 ms
 4 10.25.1.10 (10.25.1.10)  1.961 ms  1.917 ms  1.901 ms
 5 182.74.172.121 (182.74.172.121)  6.279 ms  6.272 ms  6.246 ms
 6 116.119.161.149 (116.119.161.149)  6.242 ms  116.119.161.183 (116.119.161.183)  8.452 ms
```

Figure 8: traceroute -m 15 8.8.8.8

Observation: The first hop, `samsamuel.mshome.net` (172.28.64.1), is the WSL2/Hyper-V virtual gateway on the Windows host itself – matching the default-route gateway seen in Section 2.5. From there the path crosses a couple of private-range hops (likely the ISP’s internal network / CGNAT) before reaching public IP space and, by hop 6, Google’s network, where multiple addresses answering the same hop indicate ECMP (the ISP is load-balancing across parallel links to Google’s edge). Total latency to a hop close to Google was under 10 ms, consistent with the low RTT seen from `ping`.

3.10 mtr

Purpose: Combines `ping` and `traceroute` into one continuously-updating tool – it traces the path like `traceroute` but keeps sending probes to every hop, giving a live per-hop loss/latency statistics table instead of a single `traceroute` snapshot.

Command used: `mtr -rw -c 5 8.8.8.8` (`-r` report mode / non-interactive, `-c 5` five probe cycles)

```

C:\Windows\system32\wslex x + v
sam@samsamuel:~/pcn-lab$ mtr -rw -c 5 8.8.8.8
Start: 2026-08-17T17:21:29+0000
HOST: samsamuel
  Loss%  Snt  Last  Avg  Best  Wrst  StDev
1. |-----| samsamuel.mshome.net  0.0%    5   0.1   0.5   0.1   1.0   0.3
2. |-----| 10.59.0.1                0.0%    5   1.6   1.8   1.6   2.0   0.2
3. |-----| 10.17.0.2                0.0%    5   1.6   1.7   1.5   1.9   0.2
4. |-----| 10.25.1.10                 0.0%    5   1.2   1.5   1.2   2.3   0.5
5. |-----| 122.15.35.174                0.0%    5   2.7   2.4   1.6   4.3   1.2
6. |-----| 72.14.197.130               0.0%    5   2.8   2.7   2.5   3.1   0.2
7. |-----| 192.178.121.23               0.0%    5   5.1   4.8   3.7   5.9   1.0
8. |-----| 216.239.56.65               0.0%    5   2.8   3.0   2.8   3.9   0.5
9. |-----| dns.google                  0.0%    5   4.9   3.7   2.9   5.2   1.2

sam@samsamuel:~/pcn-lab$

```

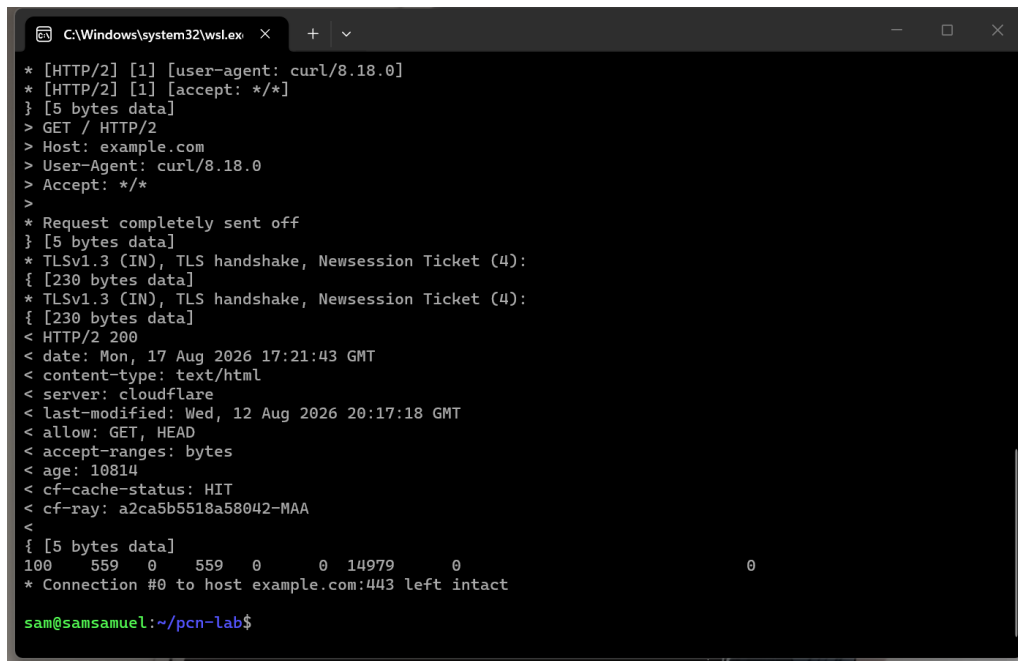
Figure 9: mtr -rw -c 5 8.8.8.8 – report mode

Observation: Across all 9 hops and 5 probe cycles each, loss stayed at 0.0% on every hop, and the Avg/Best/Worst/StDev columns show latency growing gradually with hop count, with the largest jump between the local gateway and the ISP core (hop 5, 122.15.35.174) – the point where traffic leaves the WSL/home network and enters the wider internet. The final hop resolves via reverse DNS to `dns.google`, confirming the destination is indeed one of Google’s public DNS front-ends.

3.11 curl

Purpose: Command-line tool for transferring data with a URL, supporting HTTP(S), FTP and many other protocols; `-v` prints the full request/response including the TLS handshake.

Command used: `curl -v https://example.com -o /dev/null`



```

C:\Windows\system32\cmd.exe
* [HTTP/2] [1] [user-agent: curl/8.18.0]
* [HTTP/2] [1] [accept: */*]
} [5 bytes data]
> GET / HTTP/2
> Host: example.com
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
} [5 bytes data]
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
{ [230 bytes data]
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
{ [230 bytes data]
< HTTP/2 200
< date: Mon, 17 Aug 2026 17:21:43 GMT
< content-type: text/html
< server: cloudflare
< last-modified: Wed, 12 Aug 2026 20:17:18 GMT
< allow: GET, HEAD
< accept-ranges: bytes
< age: 10814
< cf-cache-status: HIT
< cf-ray: a2ca5b5518a58042-MAA
<
{ [5 bytes data]
100  559  0  559  0  0 14979  0
* Connection #0 to host example.com:443 left intact

sam@samsamuel:~/pcn-lab$

```

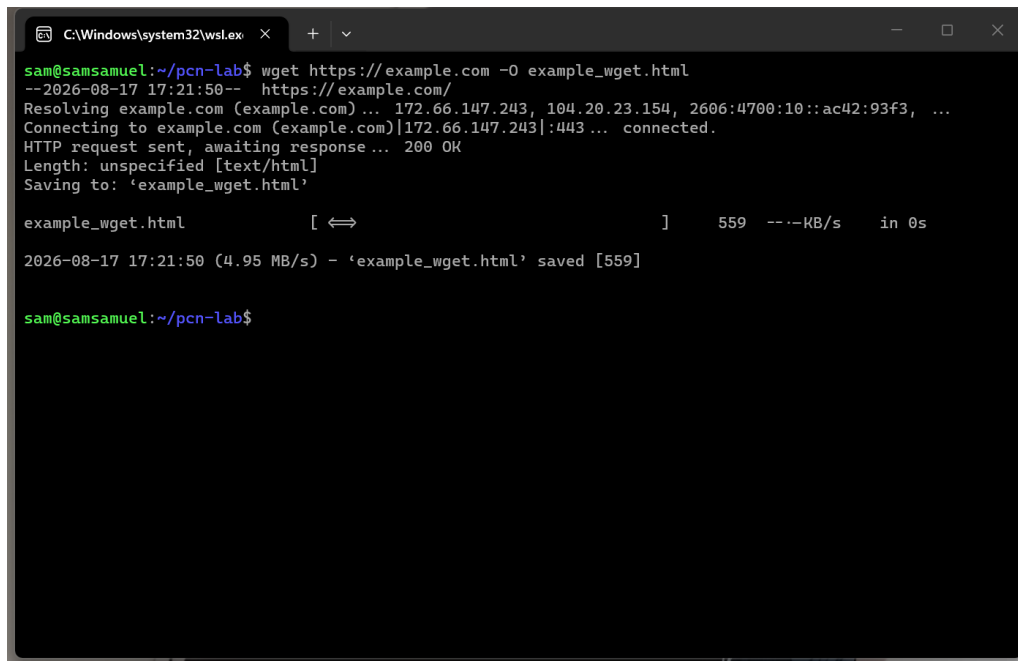
Figure 10: `curl -v https://example.com` – verbose TLS handshake and HTTP/2 response

Observation: `curl` resolved `example.com` to a Cloudflare-hosted address (`104.20.23.154`), completed a TLSv1.3 handshake (client hello, server hello, certificate, finished), negotiated HTTP/2 over ALPN, and received a 200 response served from Cloudflare’s edge cache (`cf-cache-status: HIT`). This single command demonstrates the full modern HTTPS stack – TCP connect, TLS 1.3 key exchange and certificate verification, then an HTTP/2 request/response – that Assignment 3’s C client will have to build manually at the socket level for its basic HTTP case.

3.12 wget

Purpose: Non-interactive command-line downloader, commonly used to fetch a file or whole page from an HTTP(S)/FTP URL and save it to disk.

Command used: `wget https://example.com -O example_wget.html`



```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ wget https://example.com -O example_wget.html
--2026-08-17 17:21:50-- https://example.com/
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::ac42:93f3, ...
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'example_wget.html'

example_wget.html      [ <=> ]      559  ---KB/s  in 0s

2026-08-17 17:21:50 (4.95 MB/s) - 'example_wget.html' saved [559]

sam@samsamuel:~/pcn-lab$
```

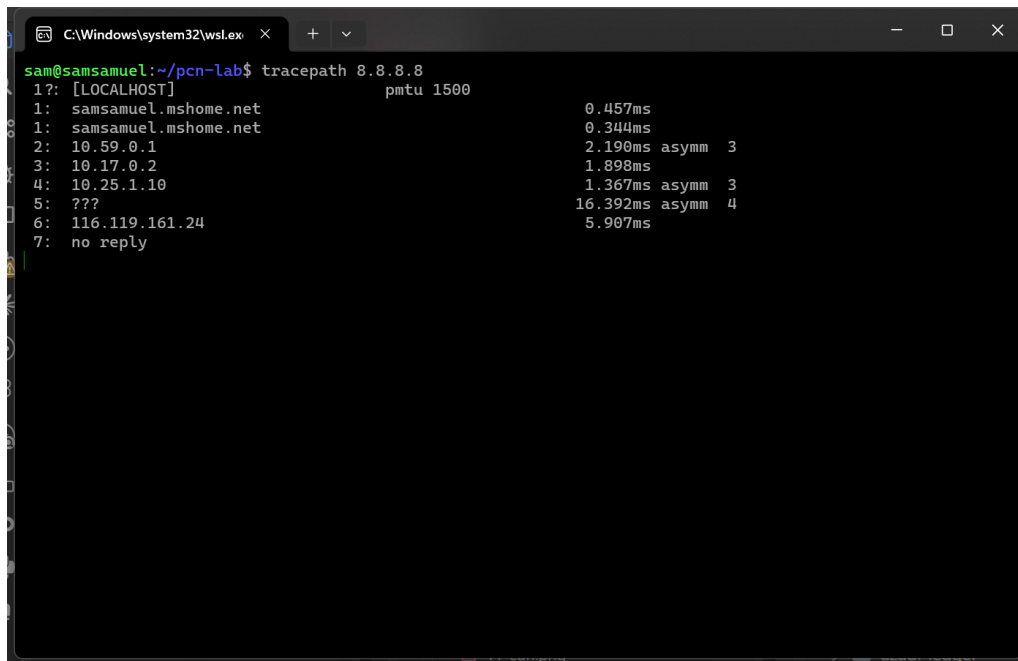
Figure 11: wget saving example.com to a local file

Observation: wget resolved and connected to the same Cloudflare address as curl, received a 200 OK with Content-Type: text/html, and saved the 559-byte response body to example_wget.html on disk – unlike curl, which by default streams to stdout, wget’s whole design centres on saving the retrieved resource to a file, which is exactly the behaviour Assignment 3’s HTTP client needs to replicate for downloading index.html / sample.pdf / image.jpg.

3.13 tracepath

Purpose: A traceroute-like tool that does not require root privileges (unlike classic traceroute), and additionally reports the discovered Path MTU.

Command used: tracepath 8.8.8.8



```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ tracert 8.8.8.8
1?: [LOCALHOST] pmtu 1500
1: samsamuel.mshome.net 0.457ms
1: samsamuel.mshome.net 0.344ms
2: 10.59.0.1 2.190ms asymm 3
3: 10.17.0.2 1.898ms
4: 10.25.1.10 1.367ms asymm 3
5: ??? 16.392ms asymm 4
6: 116.119.161.24 5.907ms
7: no reply
```

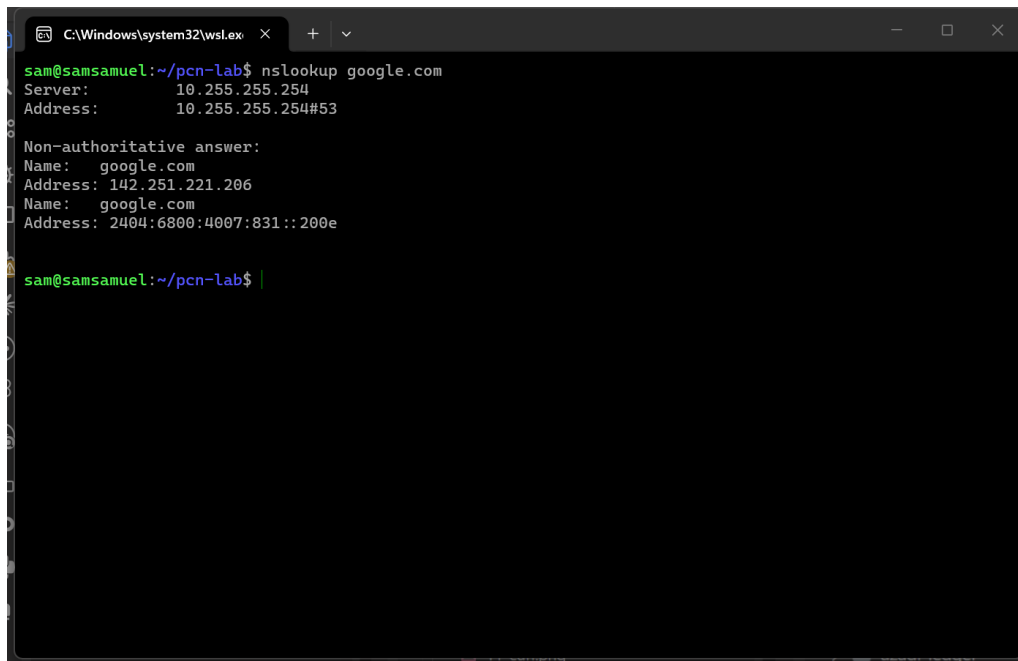
Figure 12: tracepath 8.8.8.8

Observation: tracepath discovered a Path MTU of 1500 bytes (the Ethernet standard, confirming no tunnelling/fragmentation is reducing the effective packet size along the path), and traced 6 responding hops with asymmetric round-trip marks (**asymm**) on a couple of hops – meaning the reverse path taken by the ICMP reply does not mirror the forward path exactly, which is common on the modern, multi-path internet. The final **no reply** at hop 7 is expected: many routers close to the destination rate-limit or drop the low-TTL UDP probes tracepath uses, without that indicating an actual fault.

3.14 nslookup

Purpose: Interactive/batch tool to query DNS servers for a name's resource records – the older, simpler counterpart to **dig**.

Command used: **nslookup google.com**



```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ nslookup google.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.206
Name:   google.com
Address: 2404:6800:4007:831::200e

sam@samsamuel:~/pcn-lab$
```

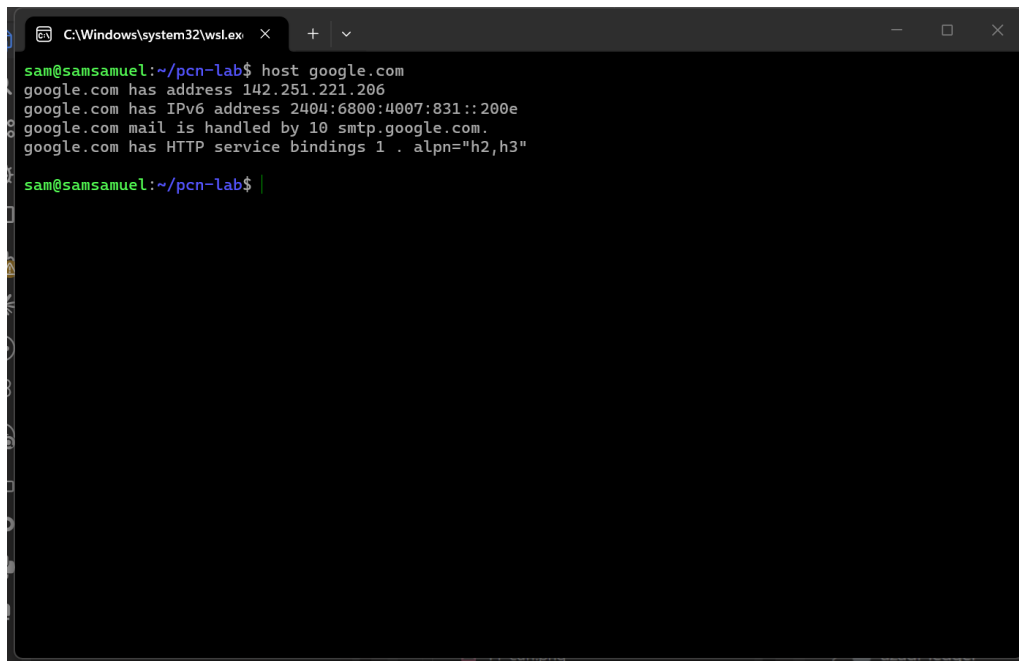
Figure 13: nslookup google.com

Observation: The resolver at 10.255.255.254 returned both an IPv4 (142.251.221.206) and an IPv6 (2404:6800:4007:831::200e) address for `google.com` under **Non-authoritative answer** – meaning the answer came from the resolver’s cache/forwarding rather than directly from Google’s authoritative nameservers, which is the normal, expected mode for an everyday DNS lookup.

3.15 host

Purpose: A concise DNS lookup utility, well suited to scripting, that by default prints A, AAAA and MX records in one line each.

Command used: `host google.com`

A screenshot of a Windows command prompt window titled 'C:\Windows\system32\cmd.exe'. The prompt shows a user named 'sam' at a host named 'samsamuel' in the directory '~/pcn-lab'. The user has entered the command 'host google.com'. The output of the command is displayed in green text: 'google.com has address 142.251.221.206', 'google.com has IPv6 address 2404:6800:4007:831::200e', 'google.com mail is handled by 10 smtp.google.com.', and 'google.com has HTTP service bindings 1 . alpn="h2,h3"'. The prompt is ready for the next command.

```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab$ host google.com
google.com has address 142.251.221.206
google.com has IPv6 address 2404:6800:4007:831::200e
google.com mail is handled by 10 smtp.google.com.
google.com has HTTP service bindings 1 . alpn="h2,h3"
sam@samsamuel:~/pcn-lab$
```

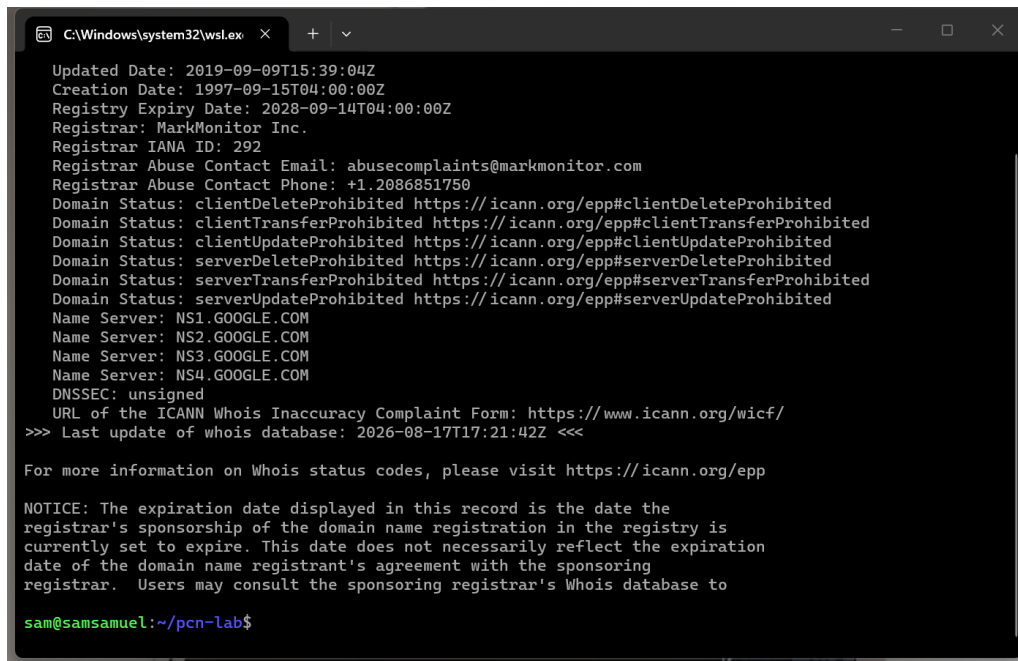
Figure 14: host google.com

Observation: In one compact query, `host` listed google.com’s A record (142.251.221.206), AAAA record, mail-exchanger record (10 smtp.google.com), and an HTTPS service-binding record advertising ALPN support for h2 and h3 (HTTP/2 and HTTP/3) – the same underlying data as `dig/nslookup`, just formatted for quick reading rather than full protocol detail.

3.16 whois

Purpose: Queries domain (or IP) registry databases for registration information – registrar, registration/expiry dates, nameservers and domain status.

Command used: `whois google.com`



```
C:\Windows\system32\cmd.exe
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGL.COM
Name Server: NS2.GOOGL.COM
Name Server: NS3.GOOGL.COM
Name Server: NS4.GOOGL.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-17T17:21:42Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
sam@samsamuel:~/pcn-lab$
```

Figure 15: whois google.com

Observation: google.com is registered through **MarkMonitor Inc.**, created **1997-09-15** and set to expire **2028-09-14**, delegated to NS1--NS4.GOOGL.COM. The domain status flags (clientTransferProhibited, clientDeleteProhibited, etc.) show it is locked against unauthorised transfer/deletion – standard practice for a high-value domain. Unlike the DNS tools above, whois queries the registry/registrar database directly rather than the DNS resolution chain, so it answers “who owns this name” rather than “what address does this name point to”.

4 Part B — Socket Programming System Calls

The table below summarises the purpose, key parameters and return value of each system call, grouped by role in the client–server lifecycle. TCP/IP and UDP applications in the last column note where a call is mandatory (M), optional/common (O), or not applicable (–) for each transport.

4.1 Socket creation, addressing and lifecycle

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>socket()</code>	<code>int socket(int domain, int type, int protocol)</code> creates an endpoint for communication. <code>domain=AF_INET</code> selects IPv4; <code>type</code> is <code>SOCK_STREAM</code> for TCP or <code>SOCK_DGRAM</code> for UDP; <code>protocol=0</code> lets the kernel pick the default for that type.	New socket file descriptor, or <code>-1</code> on error.	M (both) – first call in every socket program.
<code>bind()</code>	<code>int bind(int sockfd, const struct sockaddr *addr, socklen_t addrlen)</code> associates a socket with a local IP address and port, described by a <code>struct sockaddr_in</code> (<code>sin_family</code> , <code>sin_port</code> , <code>sin_addr</code>).	0 on success, <code>-1</code> on error.	M for servers (both); optional for clients (kernel auto-assigns an ephemeral port otherwise).
<code>listen()</code>	<code>int listen(int sockfd, int backlog)</code> marks a <code>SOCK_STREAM</code> socket as passive/listening and sets <code>backlog</code> , the maximum length of the queue of pending (not-yet-accepted) connections.	0 on success, <code>-1</code> on error.	M (TCP server only) – meaningless for UDP, which is connectionless.
<code>connect()</code>	<code>int connect(int sockfd, const struct sockaddr *addr, socklen_t addrlen)</code> . For TCP, performs the three-way handshake with the given peer. For UDP, it merely records a default peer address so subsequent <code>send()/recv()</code> can be used instead of <code>sendto()/recvfrom()</code> – no packets are exchanged.	0 on success, <code>-1</code> on error.	M (TCP client); O (UDP client, to fix the peer).
<code>accept()</code>	<code>int accept(int sockfd, struct sockaddr *addr, socklen_t *addrlen)</code> blocks until a connection is pending on a listening socket, then returns a <i>new</i> socket descriptor dedicated to that one client; <code>addr</code> is filled with the client’s address.	New connected-socket descriptor, or <code>-1</code> on error.	M (TCP server only).
<code>close()</code>	<code>int close(int fd)</code> releases the file descriptor; for TCP this initiates connection teardown (FIN) once all references to the socket are closed.	0 on success, <code>-1</code> on error.	M (both) – every socket opened must eventually be closed.

4.2 Buffers and byte order

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>bzero()</code>	<code>void bzero(void *s, size_t n)</code> fills <code>n</code> bytes at <code>s</code> with zero; routinely used to clear a <code>struct sockaddr_in</code> before filling in its fields, so padding bytes don't contain garbage.	None (void).	O (both) – good practice before populating an address struct.
<code>htons()</code>	<code>uint16_t htons(uint16_t hostshort)</code> converts a 16-bit value (typically a port number) from host byte order to network (big-endian) byte order.	The value in network byte order.	M (both) – every <code>sin_port</code> assignment.
<code>htonl()</code>	<code>uint32_t htonl(uint32_t hostlong)</code> – the 32-bit equivalent of <code>htons()</code> , used for IP addresses (e.g. <code>INADDR_ANY</code>) held as a 32-bit integer.	The value in network byte order.	O (both) – needed whenever an address is built numerically rather than via <code>inet_pton</code> .
<code>ntohs()</code>	<code>uint16_t ntohs(uint16_t netshort)</code> converts a 16-bit value from network byte order back to host byte order – the inverse of <code>htons()</code> ; used when reading a port back out of a received <code>sockaddr_in</code> to print/compare it.	The value in host byte order.	O (both) – e.g. printing a peer's port from <code>getpeername()</code> .
<code>ntohl()</code>	<code>uint32_t ntohl(uint32_t netlong)</code> – the 32-bit inverse of <code>htonl()</code> .	The value in host byte order.	O (both).

4.3 Data transfer

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>read()</code>	<code>ssize_t read(int fd, void *buf, size_t count)</code> – generic POSIX I/O call; on a connected socket it behaves like <code>recv(fd, buf, count, 0)</code> .	Bytes read (≥ 0), 0 on EOF/peer closed, -1 on error.	O (TCP, connected sockets).
<code>write()</code>	<code>ssize_t write(int fd, const void *buf, size_t count)</code> – generic POSIX I/O call; on a connected socket behaves like <code>send(fd, buf, count, 0)</code> .	Bytes written, or -1 on error.	O (TCP, connected sockets).
<code>send()</code>	<code>ssize_t send(int sockfd, const void *buf, size_t len, int flags)</code> – sends data on a <i>connected</i> socket (TCP, or UDP after <code>connect()</code>); flags such as <code>MSG_DONTWAIT</code> modify behaviour.	Bytes sent, or -1 on error.	M (TCP); O (connected UDP).

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>recv()</code>	<code>ssize_t recv(int sockfd, void *buf, size_t len, int flags)</code> – receives data on a connected socket.	Bytes received, 0 if peer closed, -1 on error.	M (TCP); O (connected UDP).
<code>sendto()</code>	<code>ssize_t sendto(int sockfd, const void *buf, size_t len, int flags, const struct sockaddr *dest_addr, socklen_t addrlen)</code> – sends a datagram to an explicit destination without requiring <code>connect()</code> first (<code>dest_addr=NULL</code> on an already-connected socket falls back to plain <code>send()</code> behaviour).	Bytes sent, or -1 on error.	M (UDP, connectionless); O (TCP).
<code>recvfrom()</code>	<code>ssize_t recvfrom(int sockfd, void *buf, size_t len, int flags, struct sockaddr *src_addr, socklen_t *addrlen)</code> – receives a datagram and records who it came from in <code>src_addr</code> .	Bytes received, or -1 on error.	M (UDP, connectionless); O (TCP).

4.4 Multiplexing, configuration and name resolution

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>select()</code>	<code>int select(int nfds, fd_set *readfds, fd_set *writefds, fd_set *exceptfds, struct timeval *timeout)</code> – blocks (up to timeout) until one of the given file descriptors is ready for I/O; the classic way to wait on several sockets, or to add a timeout to a blocking <code>accept()/recv()</code> .	Number of ready descriptors, 0 on timeout, -1 on error.	O (both) – essential for a single-threaded server handling multiple clients (Assignment 2 uses <code>pthread</code> instead).
<code>setsockopt()</code>	<code>int setsockopt(int sockfd, int level, int optname, const void *optval, socklen_t optlen)</code> – sets a socket option, e.g. <code>SOL_SOCKET/SO_REUSEADDR</code> to allow immediately rebinding a port still in <code>TIME_WAIT</code> .	0 on success, -1 on error.	O (both) – almost always used on the listening/bound socket of a server.
<code>fcntl()</code>	<code>int fcntl(int fd, int cmd, ...)</code> – general file-descriptor control; <code>F_GETFL/F_SETFL</code> read/modify flags such as <code>O_NONBLOCK</code> , commonly used to switch a socket to non-blocking mode.	Depends on <code>cmd</code> (e.g. current flags for <code>F_GETFL</code>), -1 on error.	O (both).

Call	Purpose & key parameters	Returns	TCP / UDP use
<code>getpeername()</code>	<code>int getpeername(int sockfd, struct sockaddr *addr, socklen_t *addrlen)</code> – returns the address of the peer connected to <code>sockfd</code> , useful for logging/authorising a client after <code>accept()</code> .	0 on success, -1 on error.	O (TCP, connected sockets).
<code>gethostname()</code>	<code>int gethostname(char *name, size_t len)</code> – returns the local machine's hostname.	0 on success, -1 on error.	O (both) – informational/logging use.
<code>gethostbyname()</code>	<code>struct hostent *gethostbyname(const char *name)</code> – resolves a hostname to one or more IP addresses (<code>h_addr_list</code>); the classic (now legacy, superseded by <code>getaddrinfo()</code>) way for a client to turn a server name into an address before <code>connect()</code> .	Pointer to a <code>struct hostent</code> , or NULL on failure.	O (both) – client-side name resolution.
<code>gethostbyaddr()</code>	<code>struct hostent *gethostbyaddr(const void *addr, socklen_t len, int type)</code> – reverse DNS: resolves an IP address back to a hostname (a PTR lookup), the inverse of <code>gethostbyname()</code> .	Pointer to a <code>struct hostent</code> , or NULL if no PTR record.	O (both) – e.g. a server logging the resolved name of a connecting client.

4.5 Empirical demonstration

To confirm the calls above actually behave as described rather than only on paper, a small TCP client/server pair was written in C (`as1/src/server.c`, `as1/src/client.c`) that genuinely exercises all 23 calls from the two tables: the server creates, configures, binds and listens on a socket; `select()` waits for a connection; `accept()` takes it; `fcntl()`, `getpeername()` and `gethostbyaddr()` inspect the new connection; and three request/reply turns are exchanged, one turn each over `recv()/send()`, `recvfrom()/sendto()` and `read()/write()`, so every data-transfer call is genuinely used at least once, not just described. Rather than hard-coding a demo string, both programs read each message from `stdin` (via `fgets`) exactly as a person typing at the keyboard would – the excerpt below is the server side's message loop:

Listing 1: Excerpt of `as1/src/server.c` – interactive message loop

```

1 bzero(inbuf, sizeof(inbuf));
2 recv(connfd, inbuf, sizeof(inbuf) - 1, 0);
3 strip_newline(inbuf);
4 printf("them: %s\n", inbuf);
5 printf("you: ");
6 fflush(stdout);
7 fgets(outbuf, sizeof(outbuf), stdin);
8 strip_newline(outbuf);
9 printf("%s\n", outbuf);
10 send(connfd, outbuf, strlen(outbuf), 0);

```

Two terminals were run side by side: one where the server was compiled and started (`gcc -Wall -o server server.c && ./server`), and a second, separate terminal where the client was compiled and connected to it (`gcc -Wall -o client client.c && ./client 127.0.0.1`) once the server was already listening.

```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab-src$ gcc -Wall -o server server.c
sam@samsamuel:~/pcn-lab-src$ ./server < server_talk.txt
[server] running on host: samsamuel
[server] htonl(INADDR_ANY)=0  htons(5050)=47635 (network byte order)
[server] ntohl() back = 0  ntohs() back = 5050 (host byte order)
[server] listening on port 5050 ... (waiting for the client to connect)
[server] fcntl(F_GETFL) on accepted socket = 0x2 (O_NONBLOCK bit=0)
[server] getpeername() -> client connected from 127.0.0.1:36804
[server] gethostbyaddr() -> localhost

them: hi
you: hi
them: how are you, what are you doing
you: doing good, just testing this program
them: testing success, bye-bye
you: testing success, bye-bye

[server] close() called on both sockets. Server exiting.
sam@samsamuel:~/pcn-lab-src$
```

Figure 16: Server terminal – compiles, binds/listens, accepts the connection, and exchanges three messages

```
C:\Windows\system32\cmd.exe
sam@samsamuel:~/pcn-lab-src$ gcc -Wall -o client client.c
sam@samsamuel:~/pcn-lab-src$ ./client 127.0.0.1 < client_talk.txt
[client] running on host: samsamuel, connecting to 127.0.0.1
[client] gethostbyname("127.0.0.1") -> 127.0.0.1
[client] connect() succeeded on port 5050

you: hi
them: hi
you: how are you, what are you doing
them: doing good, just testing this program
you: testing success, bye-bye
them: testing success, bye-bye

[client] close() called. Client exiting.
sam@samsamuel:~/pcn-lab-src$
```

Figure 17: Client terminal – compiles, connects, and exchanges the same three messages

Observation: The byte-order conversions are visible directly in the output – `htons(5050)` prints as 47635, which is exactly 5050 with its two bytes swapped (5050 = 0x13BA, swapped

to `0xBA13 = 47635`), and `ntohs()` converts it straight back to 5050, confirming the two functions are true inverses on this little-endian host. `getpeername()` correctly reported the client's loopback address and ephemeral source port; `gethostbyaddr()` resolved `127.0.0.1` back to `localhost`; and `fcntl(F_GETFL)` showed the accepted socket's `O_NONBLOCK` bit as 0, i.e. still blocking, since it was never switched into non-blocking mode. Both terminals show an identical three-line conversation, confirming the byte stream sent by one side's `send()/sendto()/write()` was received intact by the other side's `recv()/recvfrom()/read()` in the same order it was sent – the in-order, reliable delivery guarantee that TCP provides and that Assignment 2's UDP path will not.

5 Learning Outcome

- Learned how to use everyday Linux networking commands (`netstat`, `ifconfig`, `ping`, `route`, `nmap`, `tcpdump`, `dig`, `traceroute`, `mtr`, `curl`, `wget`, `tracepath`, `nslookup`, `host`, `whois`) to check a machine's network status, routing, and connectivity, and to read/interpret their output.
- Understood what each socket system call (`socket`, `bind`, `listen`, `accept`, `connect`, `send/recv`, `sendto/recvfrom`, `select`, etc.) actually does, instead of just knowing their names.
- Got hands-on practice writing and running a real TCP client-server program in C, and saw byte-order conversion (`htons/ntohs`) and data delivery work correctly across two live terminals.